Executable Evidence Management for CAD-Guided Aircraft Inspection

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Abstract

Inspection evidence must remain aligned with the requirement, computation and output it supports. We develop an executable evidence interface for CAD-guided aircraft inspection that combines typed quality requests, producer dependencies, calibrated event bounds and authenticated output recovery. Event containment reuses stronger evidence, while exact-event fallback preserves the support of direct renewal. A provenance-bound repair transaction restores a corrected output without erasing its audit history. A controlled aircraft-skin study executes 3,072 outputs from 512 independently sampled scenes. All complete methods admit 41 of 48 service contexts; independent held-out simultaneous risk bounds for these services remain below 0.075. Containment reduces evidence-record evaluations by 38.10% relative to exact dependency caching. Measured stage caching reduces source-renewal CPU by 56.26%, shared by the matched dependency-based implementations. Validated repair restores 1,536 distinct eligible source/request contexts that an irreversible latch blocks. A separate 72-image path-traced study tests the unchanged front end on geometric fasteners and reflective surfaces. These experiments establish an explicit workflow for retaining, renewing and recovering engineering evidence through component and output changes.

Keywords: Executable operating contracts, Evidence reuse, Engineering change management, Risk calibration, Aircraft inspection

1. Introduction

Aircraft inspection couples an image to a location on an airframe. A useful output therefore includes both a numerical result and evidence that the acquisition and mapping satisfy the requested quality. These two products have different lifetimes. Changing an exposure setting can leave the CAD geometry intact while changing blur and feature extraction. Relaxing a mapping tolerance can make existing evidence sufficient. Correcting an incorrectly attached output can restore service, provided that the correction is tied to the right observation and producer version.

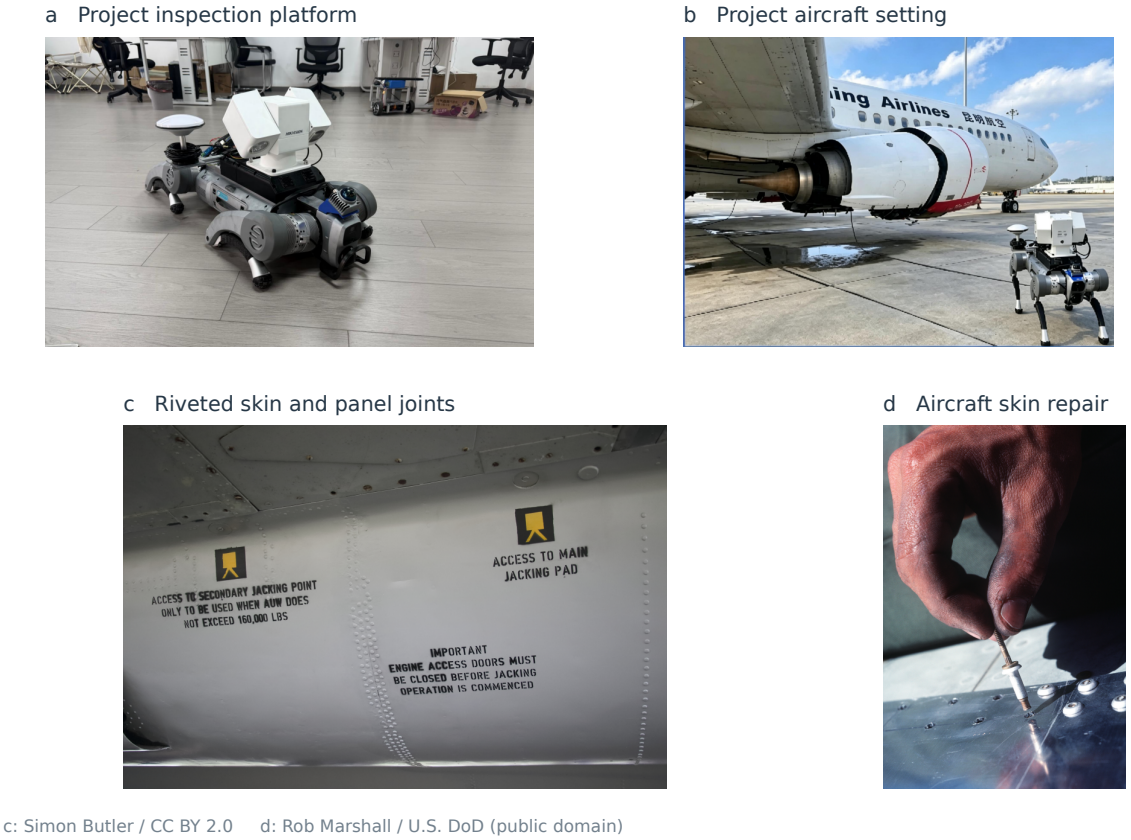
The engineering problem is deciding what remains justified after such changes. Recomputing every stage is expensive, but reusing every quality statement can attach valid evidence to an incompatible result. A dependency cache resolves numerical recomputation. The evidence decision additionally needs the meaning of the requested event, the population supporting its probability bound, and the provenance of the output receiving that statement. Treating these as an executable interface makes retention, renewal and recovery explicit operations.

Knowledge models describe engineering requirements and change dependencies [1, 2, 3]; incremental computation reuses unchanged values [4]; statistical calibration controls the risk of selected programs [5]; and runtime contracts govern permissible execution [6, 7]. We compose these mechanisms in an evidence-management workflow for CAD-guided inspection. The interface answers three questions: which statement covers the new requirement, which producer functions must agree, and which output version can receive the statement?

The paper develops three connected contributions. First, a typed request compiles engineering units and quality intervals into an event and its producer dependencies. Containment reuses stronger evidence, and exact-event fallback preserves the support of direct renewal under aligned calibration. Second, a

provenance-bound recovery protocol separates invalid output bindings from the underlying observation: a rejected attempt is retained in the audit history, while an authenticated repair can restore service. Third, an executed image benchmark links calibration, component changes, measured renewal cost and output recovery. Conventional caching, independent rules and authoritative-registry implementations identify which benefits follow from each mechanism.

The evaluation combines a controlled image population with a separate path-traced surface study. The controlled population supports repeatable calibration and paired updates; geometric rivet heads, reflective materials and area illumination in the second renderer test the image front end under a richer appearance model. Figure 1 shows the engineering context. The running example in Fig. 2 follows one observation from an exposure change to evidence renewal and verified recovery.



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Figure 1: Aircraft inspection at platform and surface scales. (a,b) Original project photographs. (c) Under-fuselage panels on Avro Vulcan XM655, photographed by Simon Butler (CC BY 2.0). (d) Aircraft sheet-metal repair, photographed by Rob Marshall (U.S. Department of Defense, public domain). Sources and image credits are archived in the Supplementary Material.

2. Related work and comparison scope

Engineering knowledge graphs connect process choices, requirements and supporting relations [1, 2]. Semantic change models propagate affected dependencies [3, 8], and inspection ontologies connect operating conditions with robot capabilities [9]. Representational completeness and executable model conversion motivate checking whether such a representation supports a concrete engineering decision [10, 11]. Here the decision is the continued applicability of evidence after a requirement, component or output-binding change.

Dependency-directed computation retains values whose inputs and functions remain unchanged [4]. A conventional dependency graph is consequently the numerical-reuse baseline. Pose estimation supplies the image-to-CAD transformation [12, 13], while uncertainty analysis distinguishes fitted residuals from transformation accuracy [14, 15, 16]. Our estimator provides the numerical outputs to which evidence is attached. A cache hit alone does not specify which quality event has a valid statistical statement.

Assumption–guarantee contracts express component obligations [6]. Binomial inversion and risk-controlled selection provide finite-family guarantees for specified programs [17, 5, 18]. Runtime assurance checks conditions for execution [7]. The present workflow connects event inclusion, requested-function identity and authenticated output provenance. Its contribution is the resulting evidence lifecycle. Equivalent rules and registry baselines test that semantics independently of the chosen software representation.

Table 1: Matched comparisons and the question each resolves.

Comparison	Shared information	Evaluated distinction
Whole renewal / DAG cache	Producer and exact events	Required numerical recomputation
DAG / contract + fallback	Same numerical cache and risk family	Event containment and evidence work
Contract / independent rules	Same event and dependency semantics	Implementation agreement
Permanent latch / validated repair	Trusted output records	Recovery after a corrected binding
Repair / authoritative registry	Same provenance and transactions	Recovery semantics across implementations

3. Inspection requirements and executable evidence

3.1. A change-driven inspection example

Consider a camera observation used to associate fastener positions with CAD addresses. The complete-service request constrains spatial sampling, motion blur, correct-address coverage, fitted residual and physical mapping error. The baseline producer satisfies the calibrated request. Increasing exposure preserves the geometric sampling scale but changes motion blur and image formation. The new producer therefore renews the affected statistical statement. A relaxed request can still be supported even when the strict event fails its risk allowance.

The numerical output also carries a source identifier, producer signature and version. If an old output is attached after the update, the binding is rejected. Correcting that binding is a separate operation from changing the quality requirement: the registry checks the active producer, output digest and repair revision before service resumes. This example keeps three decisions distinct throughout the paper: numerical reuse, statistical admissibility and output attachment. For the first held-out source, geometric sampling remains 0.858 mm/pixel after a 50% exposure increase, while blur changes from 1.050 to 1.575 pixels. This particular image satisfies the complete quality predicate in both configurations. Across the calibration population, however, the strict complete-service upper bound changes from 0.097 to 0.272, exceeding the allowance of 0.15. The relaxed exact event has upper bound 0.080 and remains admissible. Thus a good individual output and support for a producer over its operating population answer different engineering questions.

3.2. Typed requests and producer dependencies

An engineering request is $R = (g, a, \varphi, \delta)$: population g, service a, quality predicate φ and allowable scene-level failure probability δ. Its grammar is

$$\varphi ::= \mathrm{Interval}(m, u, \ell, h) \mid \varphi \wedge \varphi, \tag{1}$$

where m identifies a quantity and u its engineering unit. Compilation checks dimensions, converts units, intersects repeated bounds and canonicalizes the event. Nonfinite required values fail the quality predicate; an absent required field prevents evaluation as a schema error. The accepted set and its failure event are

$$S_R = \{z : \ell_{R,j} \le z_j \le h_{R,j},\ j \in J_R\}, \qquad E_{p,R} = \{z_p(X) \notin S_R\}. \tag{2}$$

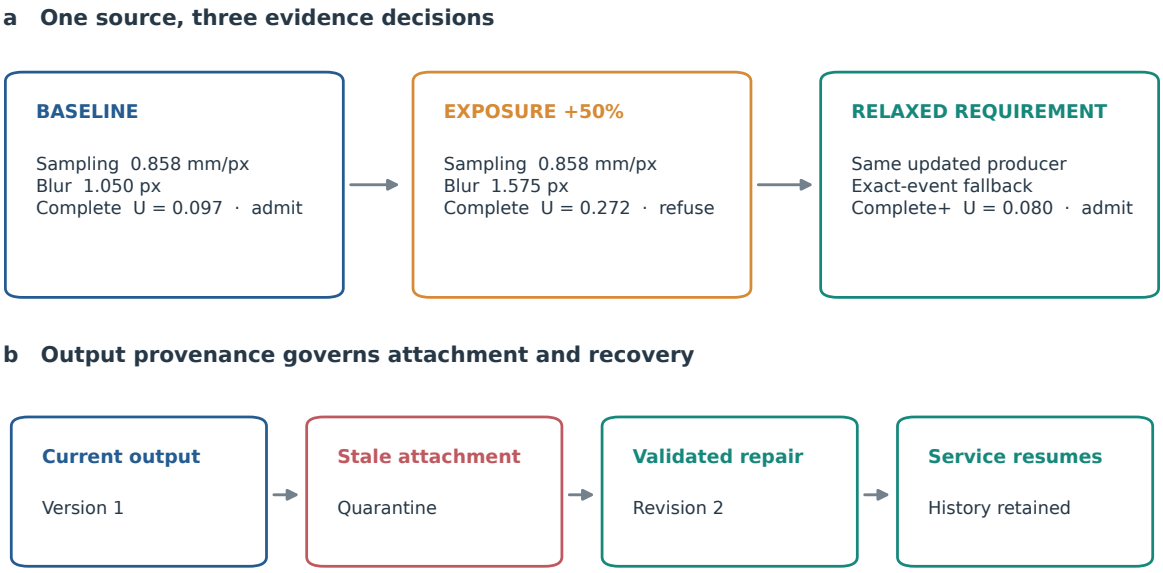


Figure 2: One inspection output through change and recovery. Exposure changes the requested blur and image quantities while preserving geometric sampling. The strict and relaxed statements use their own calibrated events. A stale attachment enters quarantine; a verified repair restores the current binding and keeps the rejected attempt in the audit record. Numerical values come from the first held-out source and the archived event cards.

The knowledge structure $\mathcal{K} = (\mathcal{V}, \tau, \mathcal{R}, \sigma, \Gamma)$ stores typed entities, typed relations, immutable specifications and executable predicates. Its relation signatures include requires : Request $\to$ Event, bounds : Card $\to$ Event, dependsOn : Quantity $\to$ Provider and uses : Receipt $\to$ SourceVersion. Type checks reject incompatible endpoints. The compiler maps these records to the decisions in Table 2.

For quantity j, the registered provider manifest declares dependencies D_j; the request uses $D_R = \bigcup_{j \in J_R} D_j$. The producer projection contains the corresponding function versions and parameters. Each numerical stage also uses an ordinary content-addressed key,

$$h_v = \mathrm{Hash}\big(v, \sigma_v, \{h_u : u \in \mathrm{pa}(v)\}\big), \tag{3}$$

which retains a cached value when the registered stage and its inputs agree. The request projection selects the functions relevant to its quality statement. A component-update regression gate re-executes common inputs and vetoes a declared invariance when an unaffected quantity changes. Runtime checks use the resulting registered signatures.

3.3. Image-to-CAD quality

Pixel extraction finds candidate fastener centers, and a pose prior supplies coarse CAD projections for correspondence assignment. For assigned correspondences $\mathcal{I}$, pose refinement solves

$$\begin{aligned} \widehat{\xi} = \arg\min_{\xi} & \sum_{i \in \mathcal{I}} \sum_{d \in \{x,y\}} \rho\left(\frac{\pi_\xi(q_i)_d - u_{i,d}}{\sigma_u}\right) \\ & + \sum_{d=1}^{3} \left[\rho\left(\frac{\Delta t_d}{\sigma_t}\right) + \rho\left(\frac{\Delta \theta_d}{\sigma_\theta}\right)\right]. \end{aligned} \tag{4}$$

Table 2: Records connecting a quality requirement to an output.

Record	Required content	Decision
Request	Population, service, units, intervals, allowance	Compile event and dependencies
Evidence card	Event, producer, calibration lineage, n, k, U	Cover the requested event
Source version	Observation identity, active producer, output digest	Authenticate the current binding
Repair transaction	Expected revision, trusted output and new revision	Restore a quarantined binding
Execution record	Plan, attachment and transaction history	Explain issuance or refusal

Here q_i is a nominal CAD address, u_i a pixel-derived center, and π_ξ the perspective projection. Translation and local rotation perturb the prior. The componentwise Huber loss has transition 2. Matching and fitting use the pixels, nominal CAD and pose prior.

Offline scoring distinguishes fitted residual from physical mapping error. Residual includes every assigned correspondence, including incorrect assignments. Mapping error evaluates the estimated image-to-CAD map $H_{\widehat{\xi}}$ at every planned physical image location:

$$e_{\mathrm{CAD}} = \max_{i \in \mathcal{P}} \|H_{\widehat{\xi}}(u_i^\star) - q_i^\star\|. \tag{5}$$

Scene geometry supplies these evaluation labels after estimation. The attachment gate authenticates output identity and provenance; physical quality labels enter calibration and evaluation.

4. Statistical reuse and validated recovery

4.1. Validity of selected services

For k failures among n independent complete calibration scenes, a one-sided Clopper–Pearson bound is

$$U(k, n; \alpha) = \begin{cases} 1, & k = n, \\ \mathrm{Beta}^{-1}(1 - \alpha; k + 1, n - k), & k < n. \end{cases} \tag{6}$$

The controlled study fixes six producers and eight events before sampling, with $K = 48$, $\beta = 0.05$ and $\alpha = \beta/K$. The common confidence event covers all 48 risks. A plan selected from these bounds is committed before evaluating a new scene.

A stronger requirement provides evidence for a weaker one: $S_A \subseteq S_R$ implies $E_{p,R} \subseteq E_{p,A}$. Producer transfer additionally requires identity of the requested functions,

$$z_{p',j}(x) = z_{p,j}(x) \quad \text{for all } j \in J_R \text{ and supported } x. \tag{7}$$

Registered dependency projections implement this condition through the declared provider interface. The numerical regression gate tests that interface against paired executions.

Proposition 1 (Validity of calibration-selected services)**.** *Let $C \sim P^n$ be calibration data and let a new scene $X \sim P$ be independent of C. Fix K producer/event functions with one-sided allocations summing to at most β. With probability at least $1 - \beta$ over C, every admitted request selected using C has fresh-scene risk at most δ, provided that its event is pointwise contained in an anchor event from this fixed family with $U_A(C) \le \delta$ and its transferred functions satisfy Eq. (7).*

Proof. The union bound gives a calibration event of probability at least $1 - \beta$ on which every fixed-family risk is bounded by its $U_A(C)$. On that event, pointwise inclusion gives $\Pr_X(E_R \mid C) \le \Pr_X(E_A \mid C) \le U_A(C) \le \delta$ for every eligible selection. Requested-function identity preserves the inclusion after transfer. □

This proposition states the quality guarantee for a planned service. Runtime attachment supplies the provenance of the result using that service. For any subsequent issuance indicator I, the joint event satisfies $\Pr(E_R \cap \{I = 1\} \mid C) \le \delta$ on the same calibration event. The Supplementary Material gives the distinction between this bound and a risk conditioned on a scene-dependent selection gate.

4.2. Containment with exact-event fallback

For a committed producer, the interface first searches compatible covering cards with $U_A \le \delta$. If none qualifies, it constructs the request's exact-event card from the fixed family. The route is

$$Q(R) = \begin{cases} \text{reuse}(A), & \mathcal{A}(R) \neq \varnothing, \\ \text{exact}(R), & \mathcal{A}(R) = \varnothing,\ U_R \le \delta, \\ \text{refuse}, & \text{otherwise.} \end{cases} \tag{8}$$

Here $\mathcal{A}(R)$ contains cards with matching population, service, producer projection and containing events whose bounds satisfy $U_A \le \delta$. Failed anchors and refusal cards count toward construction work.

Proposition 2 (Support preservation with exact fallback)**.** *Suppose the compared cards use the same n calibration scenes and allocation α, and transferred request quantities agree on those scenes. Containment followed by exact-event fallback admits exactly the requests admitted by the complete exact-event catalogue.*

Proof. Containment gives $k_R \le k_A$. At fixed n and α, the binomial upper bound is nondecreasing in k, so admission by an anchor implies $U_R \le U_A \le \delta$. Otherwise, fallback evaluates the same exact event as the complete catalogue. □

The fixed family reserves the allocation for every exact fallback. A failed strict anchor can therefore lead to an admissible relaxed event without changing the confidence budget.

4.3. Quarantine and authenticated repair

An output binding identifies a source, producer, version and payload digest. An authoritative registry stores the active tuple. Issuance requires both an admissible plan and agreement with that tuple. An invalid attachment places its binding workflow in quarantine and records the rejected attempt. Ordinary retries preserve that state.

Recovery uses an explicit transaction. It presents the expected registry revision, identifies the trusted active output, and requests a strictly newer revision. The registry verifies the producer, payload digest, source lineage and current calibration identity before accepting the transaction. A successful repair replaces the binding while retaining the audit history. A missing repair certificate, a wrong digest, a replayed transaction or an unregistered alias cannot create a valid repair. Invalid repair commands are rejected without overwriting a previously valid committed state.

The resulting invariant is direct: every issued binding equals a currently authenticated registry record, and every transition out of quarantine has a successful repair transaction in its history. Numerical quality and binding integrity have separate roles. A valid repair restores the association to a trusted output; the selected event card determines whether that producer supports the request. The comparison includes an authoritative-snapshot implementation and an independent event-log implementation, together with the earlier permanent latch. These controls quantify recoverability and check the same trust conditions through different implementations.

5. Experimental design

5.1. Controlled scene population and service requirements

The controlled image study draws 256 calibration and 256 held-out scenes from four layouts: parallel rows, curved rows, staggered fasteners and an access-panel perimeter. Each complete scene independently draws surface curvature, camera pose, material appearance and site perturbations. Layouts contain 28–36 sites, with fastener radii of 2.0–3.4 mm and typical spacing of 28–40 mm. Missing sites occur with probability 0.025; an edge occluder covers 0–10% of the raster. Shared and local physical-to-CAD displacement standard deviations are 0.4 and 0.15 mm.

Baseline resolution is 768×576 pixels. Focal length is 1,200–1,600 pixels, distance 0.70–1.15 m, exposure 3–7 ms and motion 0.06–0.18 m/s. Five changes independently increase focal length by 15%, exposure by

50%, expand the raster to 896×672, multiply illumination by 0.45, or reduce the decoder contrast threshold from 18 to 10. All six configurations execute on every scene, producing 3,072 numerical outputs. Local contrast and 5–240-pixel connected components supply candidate centers. A 30-pixel prior gate constrains minimum-cost CAD assignment; at least six matches trigger fitting with scales of 0.5 pixel, 3 mm and 0.001 rad.

Table 3 specifies the quality events used in both calibration and evaluation. These inherited benchmark requirements were fixed before sampling the new population. They describe capture-to-CAD quality at the modeled fastener scale. The primary allowance $\delta = 0.15$ applies to violation of the requested conjunction by a complete scene. A sweep from 0.02 to 0.20 reports how admission changes with this operating choice.

Table 3: Predefined capture-to-CAD requests. GSD is in mm/pixel, blur and residual in pixels, coverage in percent and mapping in mm. A dash denotes an unconstrained quantity.

Request	GSD $\leq$	Blur $\leq$	Coverage $\geq$	Residual $\leq$	Mapping $\leq$
Optical	1.2	3	—	—	—
Optical+	1.5	4	—	—	—
Acquisition	1	2	80	—	—
Acquisition+	1.2	3	60	—	—
Association	—	—	80	8	8
Association+	—	—	60	12	12
Complete	1	2	80	8	8
Complete+	1.2	3	60	12	12

A separate 128-scene stress population increases missingness to 0.14, occluded area to 0–35%, shared CAD displacement to 2 mm and local displacement to 0.6 mm. Its distribution identity remains separate from the calibrated population. The first held-out controlled input and independently rendered surfaces appear in Fig. 3.

5.2. Geometric rendering and front-end transfer

A separate Blender 4.3.2 Cycles study generates 24 independently seeded panels, each with baseline, exposure-gain and reduced-illumination configurations. Curved panel meshes, overlapping joints, shallow three-dimensional fastener heads and countersink rings form the surface. Painted and metallic materials use sampled roughness and surface variation; three area lights provide diffuse and reflected illumination. Four development seeds establish rendering settings before the held-out sources are generated.

The renderer produces 768×576 images with at most 32 samples per pixel, adaptive sampling threshold 0.01 and no denoising. The gain update adds 1.3 stops; the illumination update scales light and world radiance by 0.1. The camera is static, so this gain intervention changes brightness without a shutter-motion model. All 72 images pass through the unchanged contrast detector and pose estimator. An independent Blender camera projection is checked against the numerical projector. Physical sites enter evaluation after matching and fitting. This renderer defines a separate transfer population; its quality outcomes are evaluated directly.

5.3. Matched renewal and independent risk checks

Whole renewal recomputes the full producer and builds exact-event cards. The DAG baseline reuses unchanged stages and identical events. The contract uses the same DAG with containment and exact fallback. Independent rules implement the same interval, producer and bound predicates. All methods share the numerical estimator, calibration inputs and confidence allocation. Catalogue construction includes every evaluated event, including unsuccessful anchors.

Source-renewal timing uses 16 independent inputs, five updates and eight requests, giving 640 paired cases per method. Each case supplies the same baseline cache, rotates method order and checks numerical agreement. Process CPU and elapsed time include required stages and the scalar event check. Catalogue

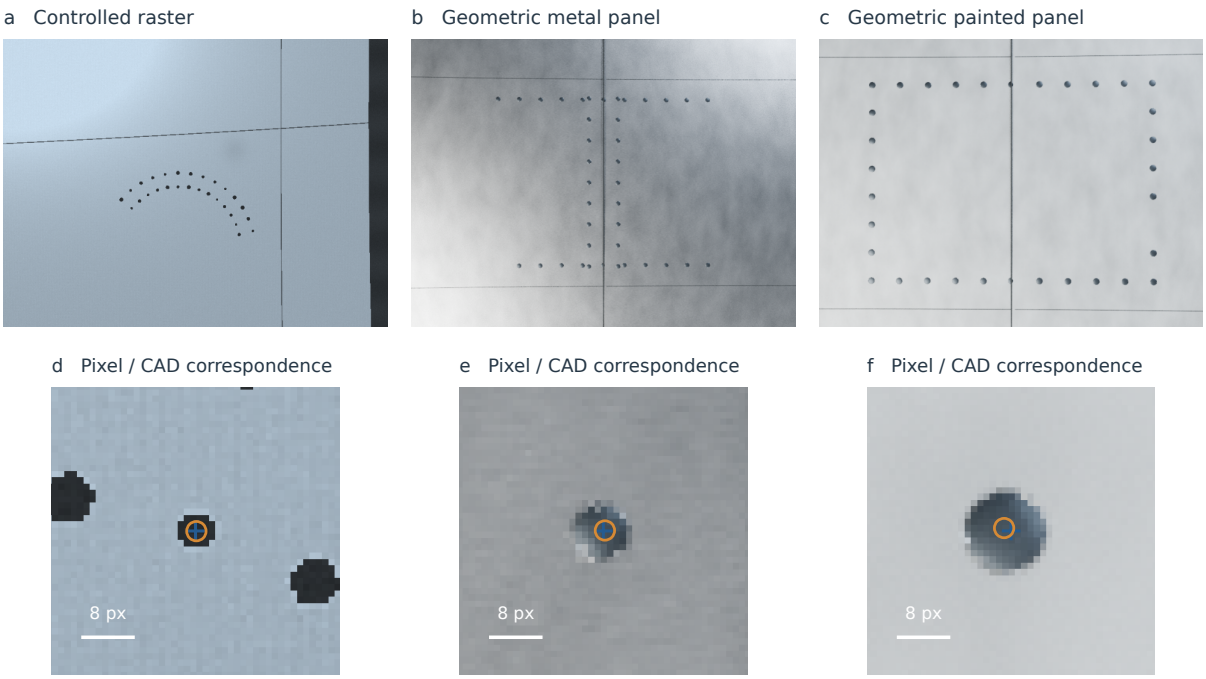


Figure 3: Images actually processed by the two producers. Controlled raster inputs support the large paired calibration study. Independently generated path-traced panels use geometric fastener heads, reflective materials and area illumination. The overlays show projected CAD locations and centers extracted from image pixels. Panels follow fixed source-order selection; their source identities and full-resolution images are archived.

preparation is measured separately over the 256 stored calibration records in 11 repetitions. Execution uses Python 3.12 on a Linux AMD EPYC 9V74 host with one numerical thread per worker.

Plans are fixed using the calibration split. Independent held-out risk checks then apply the same 48-event allocation to the test failure counts; these bounds assess the planned services. The offline component regression study uses 64 additional inputs and three conditional hidden couplings: ROI/focal length, illumination/exposure and decoder/focal length. Clean updates execute on the same inputs.

5.4. Recovery and independently checked output bindings

The recovery study replays the 256 held-out image sources under eight requests and two binding faults, yielding 2,048 distinct source/request contexts and 4,096 fault scenarios. One fault attaches a stored baseline output after the exposure update; the other attaches another source's stored output. All substituted records differ numerically from the intended output. The authenticated payload contains the archive locator, fitted residual and matched/planned counts. Physical mapping and coverage labels are reserved for independent scoring.

Each scenario follows 13 steps: baseline and updated registration, wrong attachment, retry, unregistered alias, five malformed repair attempts, valid repair, duplicate repair and resumed request. Repair evidence checks the expected revision, active producer, source, payload digest and calibration identity. The comparator derives numerical origin from the immutable archive records. Permanent latch, authoritative snapshot, validated repair, independent event-log replay and a metadata-only ablation receive the same records. Recovery availability counts only requests admitted for the updated producer.

6. Results

6.1. Calibrated services and independent held-out risk

The controlled held-out images have mean correct-address coverage 96.14%. Refinement reduces median prior translation error from 4.328 to 1.149 mm. Median fitted residual is 0.639 pixel, and median and 95th-percentile mapping errors are 0.502 and 1.052 mm.

At allowance 0.15, calibration admits 41 of 48 producer/request contexts (Fig. 4). Exposure removes strict acquisition and complete-service support; lower light additionally affects association. All complete methods give the same decisions. Anchor-only routing admits 38 contexts. Exact fallback restores relaxed acquisition and complete service after exposure, and relaxed acquisition under lower light.

The separate held-out check evaluates all 48 fixed programs on all 256 test scenes. Among the 41 calibration-admitted services, the largest test failure count is 7/256. Their simultaneous one-sided test bounds are all at most 0.07457, using a separate 95% family statement with allocation 0.05/48. These checks use the complete test population and leave the selected plans unchanged.

The allowance sweep admits 0, 18, 40, 41 and 43 contexts at 0.02, 0.05, 0.10, 0.15 and 0.20, respectively. Every complete method preserves this support. The Supplementary Material reports the corresponding construction work and all held-out bounds.

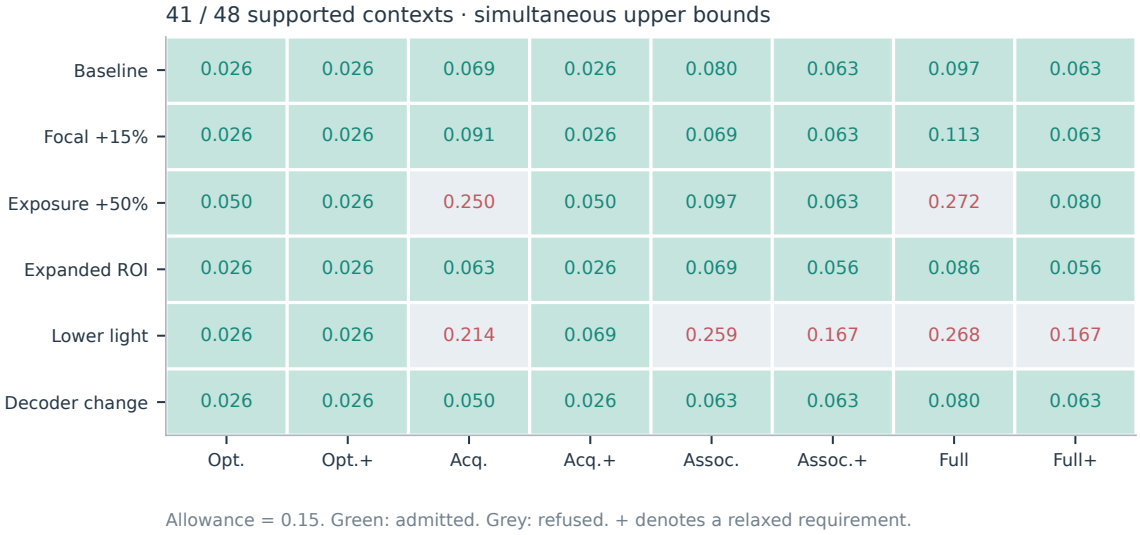


Figure 4: Calibrated event bounds for the controlled population. Green cells satisfy the primary allowance 0.15. The independent held-out check evaluates the same fixed programs after plan selection.

6.2. Renewal work and its attribution

Whole renewal constructs 48 cards; exact dependency caching constructs 42; containment with fallback and the independent rule implementation each construct 26. Their record evaluations are 12,288, 10,752 and 6,656, respectively. Containment reduces this work by 38.10% relative to exact caching while preserving all 41 supported contexts. Catalogue CPU is 4.419 ms for whole renewal, 4.290 ms for DAG exact, 2.725 ms for the compiled contract and 4.777 ms for the interpreted rules. The contract's absolute preparation saving over exact caching is 1.565 ms at this scale.

The 640 paired source-renewal cases consume 183.993 s of process CPU under whole renewal and 80.486 s under dependency caching, a 56.26% reduction. A paired bootstrap over the 16 timing inputs gives a 95% interval of 55.87–56.65%. Contract and rule paths consume 80.279 and 79.757 s. Their numerical outputs and stage counts agree with the DAG baseline. Figure 5 separates this shared numerical benefit from evidence preparation.

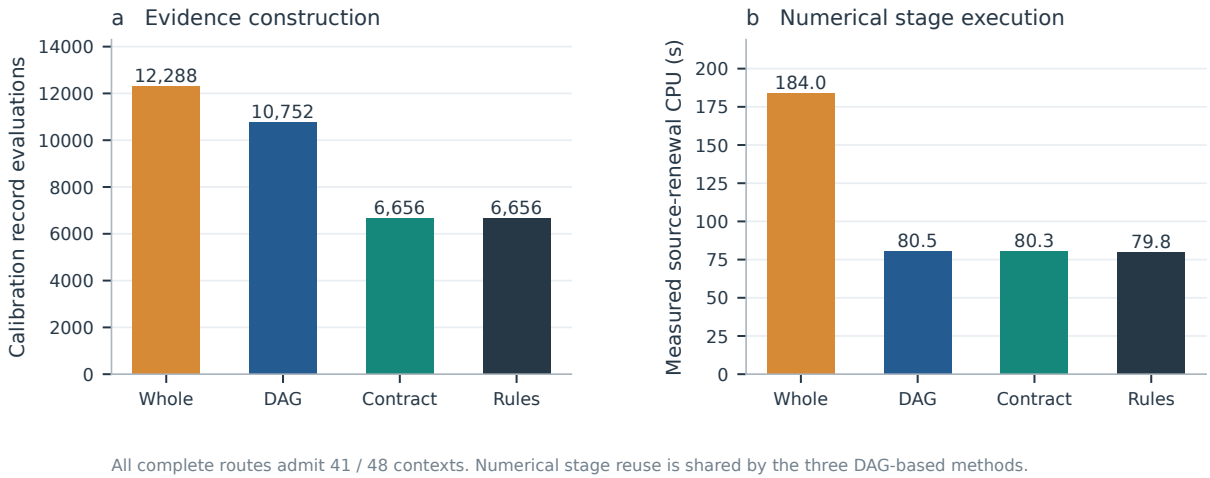


Figure 5: Two measured renewal costs. (a) Evidence construction on stored calibration records. (b) Actual numerical execution over 640 paired cases per method. Containment changes evidence work; the three dependency-based paths share numerical stage reuse.

6.3. Recoverable output attachment

Validated repair restores all 3,072 eligible fault scenarios, covering 1,536 distinct source/request contexts across the 256 images. The permanent latch restores none. The authoritative-snapshot and event-log implementations reproduce the validated-repair decisions exactly (Table 4). The source observation is retained throughout; the repair commits the trusted updated output at a newer revision.

The three recovery implementations issue no output with an incorrect numerical origin. The metadata-only ablation issues 3,072 such outputs, identifying the role of payload and provenance validation. Across all requests, 4,096 valid repairs commit and 24,576 malformed repair commands are rejected. A successfully repaired binding remains valid when a duplicate transaction is rejected. Strict complete service under exposure remains statistically inadmissible after repair; provenance recovery does not change its event bound.

Table 4: Output recovery on the same stored observations. Recovery denominator: 3,072 eligible scenarios. Wrong origin is checked against the archived source and producer record.

Method	Recovered	Wrong-origin issuances
Permanent latch	0	0
Authoritative snapshot	3,072	0
Validated repair	3,072	0
Event-log replay	3,072	0
Metadata-only ablation	3,072	3,072

6.4. Transfer to geometric surface rendering

The unchanged image front end processes all 72 path-traced images. Baseline panels have mean correct-address coverage 84.32% and median maximum mapping error 0.748 mm. At gain +1.3 stops, mean coverage is 89.54%; at one-tenth illumination it is 78.12%. The inherited association predicate fails on 9/24 baseline, 5/24 gain and 10/24 lower-light panels (Fig. 7). Full scene-level outcomes retain every rendered input.

Blender's independently computed camera projections agree with the numerical projector within 8.5×10^{-5} pixel. The resulting comparison separates appearance-driven extraction behavior from a camera-coordinate mismatch. These transfer outcomes are scored within their own renderer population; the earlier 48-event calibration remains attached to its original producer and population.

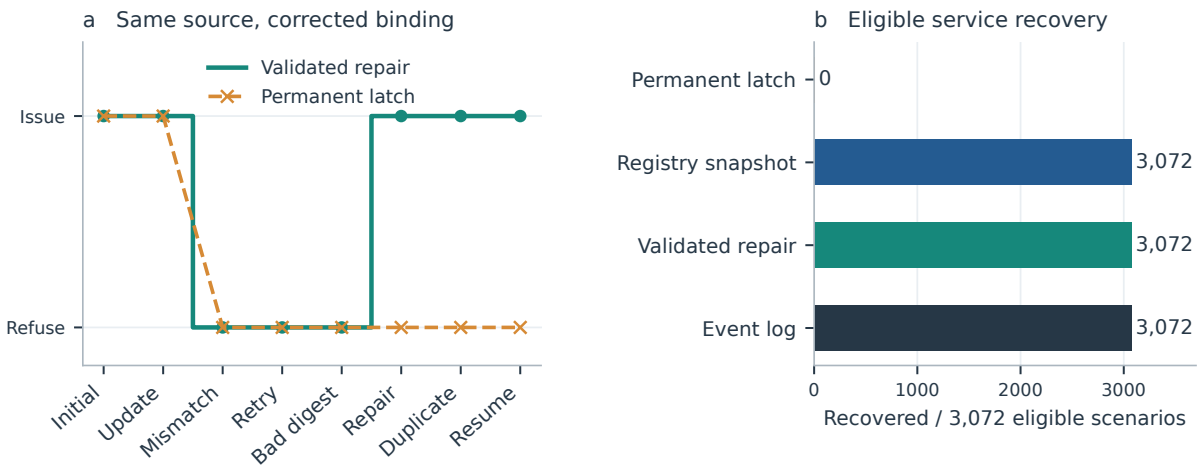


Figure 6: Recovery changes service availability while retaining the rejected attempt. (a) The first held-out source's relaxed complete-service workflow. (b) Recoverable scenarios under matched trusted-registry information. Numerical-origin checks and protocol traces use the same stored image outputs.

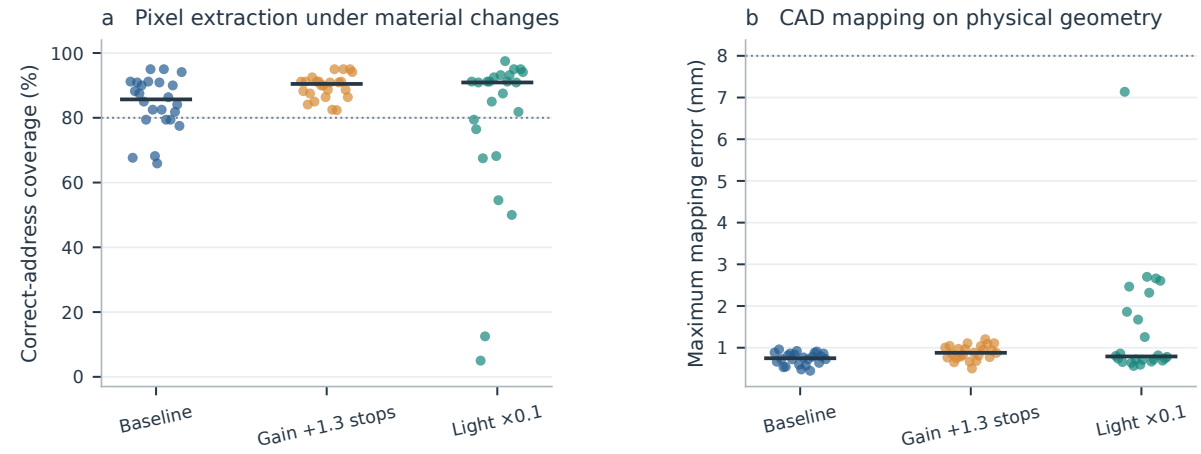


Figure 7: Image-to-CAD transfer across 24 independently rendered panels. Every scene is processed in all three configurations. The unchanged detector receives image pixels; physical CAD locations enter subsequent scoring. Dashed lines show inherited association thresholds.

6.5. Component regression and operating conditions

Conditional hidden couplings activate in 86 of 192 update/input pairs across the 64 regression inputs. Re-execution detects all 86 changes in quantities declared unaffected; all 192 clean pairs pass. The three fault classes activate on 24, 36 and 26 inputs. These checks test the declared dependency interface at component registration.

Stronger controlled disturbances lower mean coverage to 72.69% and raise the 95th-percentile mapping error to 13.095 mm. Strict complete-service failures rise from 4/256 nominal test scenes to 77/128 stress scenes (Fig. 8). Optical sampling remains acceptable, while occlusion, missing correspondences and CAD discrepancy affect the image-to-model task. This result identifies an operating-distribution change that requires its own evidence population.

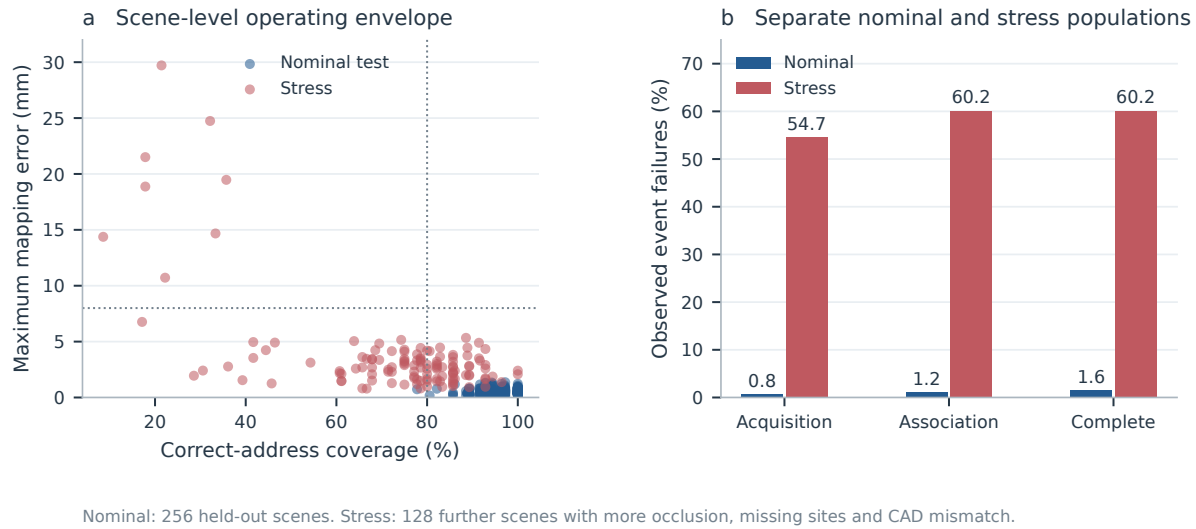


Figure 8: Controlled nominal and stress populations. Each point represents one independently sampled complete scene. Stress changes missingness, occlusion, CAD discrepancy and prior error; the resulting failures are reported separately from nominal calibration.

7. Discussion

The evidence interface resolves a change-management problem that persists after numerical caching. An unchanged stage can retain its value, a relaxed requirement can retain a stronger statistical statement, and a corrected binding can recover service without discarding the underlying observation. Each operation has a different justification. Linking them through explicit event, producer and source identities makes an inspection result explainable across its lifetime.

The comparisons locate the benefits. Ordinary stage reuse supplies the large numerical saving. Containment reduces evidence preparation while exact fallback preserves admissible requests. Authoritative registry and event-log implementations recover the same valid bindings as the contract, showing that the recovery rules are reproducible across representations. The permanent-latch comparison identifies the service availability lost when a rejected attempt is treated as an irreversible source failure.

The controlled and path-traced image studies answer complementary questions. The controlled population provides an independently sampled calibration/test experiment with known physical labels. Geometric rendering challenges extraction under a different appearance model. A change of renderer therefore creates a new population identity in the evidence system. The observed transfer behavior informs the choice of

numerical provider and calibration population rather than silently carrying the old statement into the new domain.

Engineering interpretation follows the requested event. A low fitted residual describes consistency with extracted centers; physical mapping error assesses the CAD address assigned to a known location. Population risk describes how often a specified service violates its complete quality predicate. Output integrity establishes which computation the statement accompanies. Keeping these quantities explicit prevents a successful fit or valid digest from being used as a substitute for another kind of evidence.

The workflow assumes a trusted source registry and declared producer dependencies. Registration regression exposes violations on executed inputs, while deployment of a changed provider also requires maintaining its interface. The archived manifests, transaction traces and full numerical records make these assumptions inspectable. Together, they provide a reusable engineering specification for selective renewal and recoverable evidence attachment.

8. Conclusion

Executable evidence management connects a quality requirement to the calibrated producer and authenticated output that support it. Containment with exact fallback preserves direct-renewal support, while validated repair restores a corrected binding and retains its audit history. The controlled image study supports 41 service contexts, independently checks their held-out risks and reduces evidence-record evaluations by 38.10%. Matched implementations separate this evidence benefit from conventional numerical caching and reproduce recovery across 1,536 eligible source/request contexts. The geometric-rendering and stress studies expose the role of appearance and operating-population changes. The resulting interface provides a concrete specification for selective renewal and recoverable evidence attachment in CAD-guided inspection.

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